

1. Executive Summary

INFERRED

OBSERVED THREAT LEVEL

medium

Key Observed Findings:

- Observed: ['TCP retransmission rate is at least 5 percent.']
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ATTACK HORIZON & FORECAST STATUS

Attack Horizon state is **EARLY_SIGNAL** with **60s lead time (60s duration)**.

Epistemic Policy: Epistemic Note: Observed detections reflect packet-level heuristics; Forecasts reflect multi-step world model predictions. Forecast scores are strictly uncalibrated margins, not calibrated probabilities.

2. Capture Quality

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PARSED / TOTAL

2277 / 2277

QUALITY STATUS

DEGRADED

Malformed Packets	0
Truncated Packets	35
Packet Loss Ratio	0.00%
Reordered Packets	0

3. Network Activity Summary

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TOTAL VOLUME

2,277 pkts

TEMPORAL WINDOW COVERAGE

600.0s

Packet Timestamp Span: 500.46s

Reconstructed Flows	283
Unique Host Pairs	137 src → 139 dst
Destination Ports	259
Protocols	ARP: 56, ICMPv6: 21, IPv4_PROTO_2: 4, TCP: 1617, UDP: 519, UNSUPPORTED: 60

4. Temporal Behavior

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WINDOWS (60S)

10

CONTINUITY

CONTINUOUS

RATE TREND

STABLE

FLOW CHURN

STABLE

5. Multi-Horizon Forecast Rollout & Attack Horizon

FORECAST

ATTACK HORIZON STATE

EARLY_SIGNAL

Early attack signal detected at horizon T+1 (lead time 60s); duration isolated to 60s (1 window). Temporal consistency: 1.00.

HORIZON DURATION & LEAD TIME

60s (1 windows)

Lead Time: 60.0s

Horizon	Predicted Score	Predicted Stage	Uncertainty	Top Contributing Signal
T+1 (60s)	60.02%	RECONNAISSANCE	0.80	Feature 'proto_udp_count' increasing by 410209592.8% relative to current state.; Feature 'proto_other_count' increasing by 80579679.9% relative to current state.; Inter-arrival time increasing (0.00s -> 107.64s) indicates irregular pacing or command-and-control interval changes.; Inter-arrival time increasing (0.00s -> 107.64s) indicates irregular pacing or command-and-control interval changes.; Feature 'tcp_fin_count' increasing by 1023737.4% relative to current state.
T+2 (120s)	20.85%	STABLE_BENIGN	0.42	Feature 'proto_udp_count' increasing by 371238138.5% relative to current state.; Inter-arrival time increasing (0.00s -> 419.58s) indicates irregular pacing or command-and-control interval changes.; Inter-arrival time increasing (0.00s -> 419.58s) indicates irregular pacing or command-and-control interval changes.; Feature 'proto_other_count' increasing by 34613548.4% relative to current state.; TCP flag pressure increasing (0 -> 0) reflects connection exhaustion attempts or failed handshakes.
T+3 (180s)	15.38%	STABLE_BENIGN	0.31	Feature 'proto_udp_count' increasing by 404320392.5% relative to current state.; Inter-arrival time increasing (0.00s -> 425.97s) indicates irregular pacing or command-and-control interval changes.; Inter-arrival time increasing (0.00s -> 425.97s) indicates irregular pacing or command-and-control interval changes.; Feature 'proto_other_count' increasing by 6637794.4% relative to current state.; TCP flag pressure increasing (0 -> 0) reflects connection exhaustion attempts or failed handshakes.
T+4 (240s)	11.74%	STABLE_BENIGN	0.23	Feature 'proto_udp_count' increasing by 414690620.3% relative to current state.; Inter-arrival time increasing (0.00s -> 431.00s) indicates irregular pacing or command-and-control interval changes.; Inter-arrival time increasing (0.00s -> 431.00s) indicates irregular pacing or command-and-control interval changes.; Feature 'proto_other_count' increasing by 5215494.3% relative to current state.; TCP flag pressure increasing (0 -> 0) reflects connection exhaustion attempts or failed handshakes.
T+5 (300s)	9.44%	STABLE_BENIGN	0.19	Feature 'proto_udp_count' increasing by 415294633.9% relative to current state.; Inter-arrival time increasing (0.00s -> 398.82s) indicates irregular pacing or command-and-control interval changes.; Inter-arrival time increasing (0.00s -> 398.82s) indicates irregular pacing or command-and-control interval changes.; Feature 'proto_other_count'

Horizon	Predicted Score	Predicted Stage	Uncertainty	Top Contributing Signal
				increasing by 7007517.9% relative to current state.; TCP flag pressure increasing (0 -> 0) reflects connection exhaustion attempts or failed handshakes.

Methodological Disclosure: Predicted scores represent uncalibrated model transition signals; not empirical probabilities of attack occurrence. Contextual MITRE ATT&CK mappings represent contextual behavioral alignments derived from telemetry distributions, not direct payload signature classifications.

6. Evidence Chain (Observable Corroboration)

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Evidence Strength: 0.34 Observation supported by 4 indicators, but qualified by 8 contradictory signals (e.g., Active concurrent flow volume (flow_count) decreased 95.2% during an attack forecast window.). (DEGRADED Quality)

Alignment	Feature	Observed vs Baseline	Severity	Physical Telemetry Rationale
SUPPORTING	total_src_bytes	175466.0 vs 66246.2 (+164.9%)	HIGH	Source data transmission volume (total_src_bytes) increased 164.9% over baseline (175466.0 vs 66246.2).
SUPPORTING	total_dst_bytes	89218.0 vs 24494.6 (+264.2%)	HIGH	Destination data reception volume (total_dst_bytes) increased 264.2% over baseline (89218.0 vs 24494.6).
SUPPORTING	delta_total_bytes	198022.0 vs 7406.6 (+2573.6%)	HIGH	Sequential byte volume acceleration (delta_total_bytes) increased 2573.6% over baseline (198022.0 vs 7406.6).
SUPPORTING	attack_horizon	1.0 vs 0.0 (+100.0%)	MEDIUM	Early attack signal isolated to horizon T+1.
CONTRADICTIONARY	flow_count	2.0 vs 41.3 (-95.2%)	HIGH	Active concurrent flow volume (flow_count) decreased 95.2% during an attack forecast window.
CONTRADICTIONARY	total_packets	250.0 vs 321.7 (-22.3%)	LOW	Total aggregate packet volume (total_packets) decreased 22.3% during an attack forecast window.
CONTRADICTIONARY	unique_dst_ports	2.0 vs 26.2 (-92.4%)	HIGH	Destination port diversity (unique_dst_ports) decreased 92.4% during an attack forecast window.
CONTRADICTIONARY	unique_src_ports	2.0 vs 35.9 (-94.4%)	HIGH	Source port diversity (unique_src_ports) decreased 94.4% during an attack forecast window.
CONTRADICTIONARY	proto_tcp_count	2.0 vs 29.4 (-93.2%)	HIGH	TCP protocol distribution count (proto_tcp_count) decreased 93.2% during an attack forecast window.
CONTRADICTIONARY	proto_udp_count	0.0 vs 9.9 (-100.0%)	HIGH	UDP protocol distribution count (proto_udp_count) decreased 100.0% during an attack forecast window.
CONTRADICTIONARY	delta_flow_count	-81.0 vs 9.9 (-920.2%)	HIGH	Sequential flow count acceleration (delta_flow_count) decreased 920.3% during an attack forecast window.

Alignment	Feature	Observed vs Baseline	Severity	Physical Telemetry Rationale
CONTRADICTIONARY	capture_quality	0.7 vs 1.0 (-30.0%)	MEDIUM	Underlying network capture is flagged DEGRADED.; Capture contains 1 timestamp anomalies.; Capture contains 35 truncated packets.

7. Confidence & Unknown Behavior

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Calibration Disclaimer:

Forecast score is uncalibrated and must not be interpreted as a calibrated probability.

Raw Model Score	0.6002
Calibrated Confidence	WITHHELD (UNSUPPORTED)
Uncertainty Level	HIGH

Behavior Classification:

UNKNOWN_BEHAVIOR

Conflicting evidence: contradictory signals exceed supporting indicators across multiple dimensions.

Coverage: 45.0% · Abstain Recommended: False

8. Abstention Status & Preconditions

ABSTAINED

AVAILABILITY GATE STATUS

FORECAST_AVAILABLE_BUT_UNCALIBRATED

Forecast is available and supported by lookback history, but model calibration is UNSUPPORTED. Raw scores should not be treated as calibrated probabilities.

9. Operational Limitations

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- Underlying network capture is flagged DEGRADED.
- Capture contains 1 timestamp anomalies.
- Capture contains 35 truncated packets.
- Forecast score reflects raw model margin. Calibration status is UNSUPPORTED for single-class/benchmark regimes.
- Heuristic packet alerts are indicators, not definitive intrusion ground truth.

10. Provenance & Auditability

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Source File	friday_10windows_slice.pcap (833,081 bytes)
SHA-256 Hash	N/A
Model Version	nexsolve-v1.0-research
Processing Version	nexsolve-v1.0.0-prod
Execution Time	0.00s (Stage: completed)